

Exercise 1

Let f be a map (application) from E to F . $A_1, A_2 \subset E$ and $B_1, B_2 \subset F$ Prove that :

1) $A_1 \subset A_2 \implies f(A_1) \subset f(A_2)$.

2) $f(A_1 \cap A_2) \subset f(A_1) \cap f(A_2)$.

Give a counter example to show that the inclusion is strict.

3) $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$

Exercise 2

1) Let

$$\begin{array}{ccc} f: \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^2 \end{array} \text{ and } A = [-1, 4]$$

Determine $f(A)$ and $f^{-1}(A)$.

2) We consider the sin function.

- What is the direct image by sin of $\mathbb{R}$? , of $[0, 2\pi]$? of $[0, \frac{\pi}{2}]$?

- What is the inverse (reciprocal) image by sin of $[0, 1]$? of $[3, 4]$? of $[1, 2]$?

Exercise 3

Are the following applications injective ? surjective ? bijective ?

1. $f: \mathbb{N} \rightarrow \mathbb{N}, f(n) = n = 1$.

2. $g: \mathbb{Z} \rightarrow \mathbb{N}, g(n) = n + 1$.

3. $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2, h(x, y) = (x + y, x - y)$.

Exercise 4

Let $f: \mathbb{R} \rightarrow \mathbb{R}_+^*$ dfined by $f(x) = \frac{e^x + 2}{e^{-x}}$

1. Prove that f is bijective.

2. Calculate its reciprocal bijection.

Exercise 5

I. Show that if the following relations are reflexive ?, symetric ? , antisymetric ?, transitive ?.

1) $x, y \in \mathbb{Z} : x\mathcal{R}y \Leftrightarrow x = -y$.

2) $x, y \in \mathbb{R} : x\mathcal{R}y \Leftrightarrow \cos^2 x + \sin^2 y = 1$.

3) $x, y \in \mathbb{N} : x\mathcal{R}y \Leftrightarrow y = px^q, (p, q \in \mathbb{N}^*)$.

II. Among the previous examples, which are relations of equivalence, and which are relations of order.

Exercise 6

I) Let $\mathcal{R}$ a binary relation defined on $\mathbb{N}$ by $a, b \in \mathbb{N} : a\mathcal{R}b \Leftrightarrow \exists k \in \mathbb{Z} / a - b = 3k$

1) Prove that $\mathcal{R}$ is an equivalence relation on $\mathbb{N}$.

2) Find the equivalence class $\dot{n}$ for $n = 0, 1, 2, 3$.

3) If $k \in \mathbb{N}$, is $\mathcal{R}$ an equivalence relation ? justify.

II) Show that the binary relation $\mathcal{R}$ defined on $\mathbb{R}$ by $x, y \in \mathbb{R} : x\mathcal{R}y \Leftrightarrow e^x < e^y$ is an order relation.